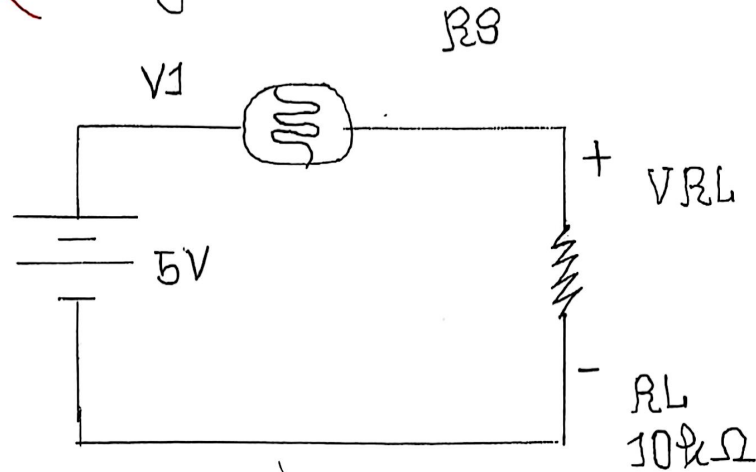


□ Math (MQ sensor)



R_S depends on gas volume

$R_S \uparrow$ amount of gas $\uparrow$ $V_{RS} \uparrow$

$R_S \downarrow$ amount of gas $\downarrow$ $V_{RS} \downarrow$

* ADC Analog to Digital Converter

1 bit	can generate	2 numbers	2^1
2 bit	" "	4 "	2^2
3 "	" "	8 "	2^3

⇒ Arduino uses 10 bit to convert analog to digital.

10 bit can generate $2^{10} = 1024$ numbers
 $\downarrow$
 (0 - 1023)

Maximum number Arduino can generate
 $= 1023$

Maximum voltage drop gas sensor can sense
 $= 5V$

$$\begin{array}{rclcl} 5V & \text{maps with} & 1023 \\ 1V & \parallel & \parallel & \frac{1023}{5} \\ xV & \parallel & \parallel & \frac{1023x}{5} = y \end{array}$$

* Digital number means discrete numbers
(0 1 2 3 4 5 6 7)

$$\frac{1023x}{5} = y$$
$$\Rightarrow x = \frac{5y}{1023}$$

x = Measured voltage dropped across the sensor (V) $= V_{RL}$

y = Digital value read by Arduino's 10 bit analog to digital converter
 $= \text{Sensor Value}$

R_S = Internal resistance of sensor

R_L = Load resistor.

V_{RL} = Voltage across load resistor.

$$V_{RL} = \frac{R_L}{R_L + R_S} \times 5$$

R_0 = Sensor resistance in clean air

Measure relative change in
sensor resistance

$$\text{ratio} = \frac{R_S}{R_0}$$

in 1 million how many gas
molecules are there

$$\text{ppm} = 10^{\left(\frac{\log_{10}(\text{ratio}) - b}{m} \right)}$$

b = calibration constant
 m = calibration slope

Problem (1)

An Arduino microcontroller with a 12-bit analog-to-digital converter is connected to an MQ-9 gas sensor. The system operates with a 5V power supply. The sensor's internal resistance (R_G) is $15k\Omega$ in clean air ($R_0 = 15k\Omega$) and load resistance (R_L) is $10k\Omega$. When exposed to carbon monoxide the internal resistance drops to $8k\Omega$. The ADC reads a digital value of 2048.

- 1) Calculate voltage dropped across load resistor.
- 2) Calculate the relative resistance ratio
- 3) Calculate maximum number of discrete distance voltage levels this 12 bit system can discriminate, ppm=?

(1)

Given,

Internal resistance of sensor, $R_G/R_0 = 15k\Omega$
 $R_G = 15k\Omega$

Internal resistance of sensor after getting exposed to CO, $R_G = R_S = 8k\Omega$

Load resistance, $R_L = 10k\Omega$

$$\begin{aligned} V_{RL} &= \frac{R_L}{R_L + R_S} \times 5 \\ &= \frac{10}{10+8} \times 5 \\ &= 2.78V \end{aligned}$$

Different approach:

Given,

Digital value read by Arduino 12 bit
analog to digital converter (ADC), $N = 2048$

12 bit can generate $2^{12} = 4096$ numbers
 $= (0 - 4095)$

Maximum number Arduino can generate
 $= 4095$

Maximum voltage drop given sensor can show
 $= 5V$

5V can map	4095
1V can map	$4095/5$
xV can map	$\frac{4095x}{5} = N$

$N = 2048 = \frac{4095x}{5}$
$\Rightarrow 4095x = 5 \times 2048$
$\Rightarrow x = \frac{5 \times 2048}{4095} = 2.5V$

(2)

Clean air resistance, $R_0 = 15 k\Omega$

$$\begin{aligned} \text{ratio} &= \frac{R_s}{R_0} \\ &= \frac{8}{15} \\ &= 0.533 \end{aligned}$$

(3)

Maximum disperse level = 4095

$$\begin{aligned} \text{ppm} &= 10^{\frac{\log_{10}(\text{ratio}) - b}{a m}} \\ &= 10^{\frac{\log_{10}(0.533) - 0.326}{-0.824}} \\ &= 10^{0.727} \\ &\approx 5.33 \text{ ppm CO} \end{aligned}$$

b & m different for different gas,